MILITARY SPECIFICATION 6 August 1971

TEST SET, RADIO FREQUENCY POWER AN/USM-161( )

1. SCOPE

1.1 This specification covers one type of radio frequency power measuring equipment designated as Test Set, Radio Frequency Power AN/USM-161( ) which consists of the following:

Test Set, RF Power	TS-1776( )/USM-161
Attenuator, Fixed	CN-844( )/USM-161
Attenuator, Fixed	CN-845( )/USM-161
Bolometer, RF	DT-255( )/USM-161
Cable Assembly, RF	CG-2514( )/USM-161

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect at date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
NN-P-71	Pallet, Materials-handling (Wood)
QQ-S-781	Steel, Strapping, Flat
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-wall
PPP-F-320	Fiber; Corrugated and Solid, Steel Stock (Container Grade), and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic, (and Connector)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MIL-E-75	Electron, Tube Packing, and Container Marking for General Specification for
MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads; 40 in x 48 in 4-way (Partial) Pallet Skids, Runners, or Pallet-type Base
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	*List of Accessories for Package Tester*
SC-DL-429477	Test Set, RF Power AN/USM-161( )
SM-C-429576	Test Procedures, Bolometer

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

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3. REQUIREMENTS

3.1 First article samples. The contractor shall furnish three (3) complete First Article samples of the test set for approval provided the invitation for bid and contract requires First Article samples.

3.2 Construction. The test set shall be constructed in accordance with SC-DL-429477 and requirements specified herein, with Equipment Verification Review (EVR) in accordance with 4.11.

3.3 Parts, materials and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection of requirements, therein, shall apply.

3.4 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware, such as screws, etc., and necessary touch-up after mounting.

3.5 Marking. Marking shall conform to MIL-M-13231.

3.5.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with a minimum disassembly of the equipment.

3.5.2 Serial numbers. Serial numbers are required on each nomenclatured (item name and type designation) item, except cords and cables.

3.6 Preconditioning.

3.6.1 Bounce preconditioning. Each equipment produced shall be subjected to the bounce preconditioning of 4.6.1. After bounce preconditioning the equipment shall be inspected for loose or broken parts and controls. If a failure occurs, the loose or broken part or control shall be repaired or replaced and the test repeated. Bounce preconditioning shall be performed prior to electrical preconditioning and Group A inspection.

3.6.2 Electrical preconditioning. Each equipment produced shall be subjected to the electrical preconditioning of 4.6.2. Electrical preconditioning shall be performed after bounce preconditioning and prior to Group A inspection. The equipment shall be monitored for measurement of 0.3 mW $\pm$ 5 percent of power at a frequency between 10 MHz and 1 GHz during period of test.

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3.7 Electrical.

3.7.1 Power requirements. The instrument shall be capable of operating within specifications with line voltages of 115V ±10 percent, between any line voltage frequency between 50 and 400 Hz. When tested in accordance with 4.7.1.1, the equipment shall meet the following requirements:

3.7.1.1 Power supply.

a. The B negative voltage shall be 228 Vdc ±5 percent.

b. The B negative voltage regulation shall be such that when the line voltage is varied from 105 to 129V, with the line voltage frequency of first 60 and then 400 Hz, the B negative voltage shall not vary by more than ±0.5 volts.

c. The B negative ripple shall be less than 25 millivolts, peak to peak.

3.7.1.2 Bias supply. Gain, calculated by dividing horizontal deflection voltage (approximately 14 volts, peak to peak) by vertical deflection voltage (approximately 150 millivolts, peak to peak) shall be greater than 50 when tested in accordance with 4.7.1.2.

3.7.2 Calibration.

3.7.2.1 Constant current supply. When calibrated in accordance with 4.7.2.1 the indicated voltages shall be within the limits specified below:

Comp Attenuator (dB)	Nominal Voltages (Vdc)	Limits of Voltages (Vdc)
+2.2	1.9428	1.9234-1.9622
0	2.5028	----
-2.2	3.2242	3.1590-3.2880

Power Range (mW)		
0.3	1.40745	1.4004-1.4145
1.0	2.50280	----
3.0	4.45072	4.4285-4.4730
10.0	7.91466	7.8750-7.9542

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3.7.2.2 Automatic zero adjustment. When calibrated in accordance with 4.7.2.2 the voltage E 1 measured across C22 shall be equal, whether the BIAS-READ switch is set to BIAS or READ.

3.7.2.3 Maximum power calibration. When calibrated in accordance with 4.7.2.3, the voltage E3 shall be equal to 0.15645 of $E_1 - E_2$.

3.7.2.4 Minimum power calibration. When calibrated in accordance with 4.7.2.4, the voltage E_4 shall be equal to 0.57822 of $E_1 - E_2$.

3.7.3 Calibration verification. The following will verify the proper calibration of the equipment.

3.7.3.1 Power meter. When tested in accordance with the procedures of 4.7.3.1, the results shall be within ± 1 percent of 1.00.

3.7.3.2 Comp attenuator. With COMP ATTENUATOR control set to -2.2 db, the indication on the voltmeter shall be within ±1 percent of 1.288 times the indication in step f. of 4.7.3.1 and with COMP ATTENUATOR control set to +2.2 db the indicator on the voltmeter shall be within ±1 percent of 0.7763 times the indication in step f. of 4.7.3.1, when tested per 4.7.3.2.

3.7.4 Operational. When tested in accordance with the procedures of 4.7.4, the equipment shall meet the following requirements:

3.7.4.1 Sensitivity of null indication. Sensitivity shall be between 0.005 and 0.02 mW.

3.7.4.2 Power measurement. Results shall meet the accuracy requirements of 3.7.5 when rf power of 0.3, 1 and 3 mW, at 1 GHz, is applied to the test set.

3.7.4.3 Zero shift. Zero shift shall not exceed 0.003 mW.

3.7.5 Accuracy of power measurements. When measured in accordance with 4.7.5, taking into consideration losses due to reference power errors of ±1/2 of 1 percent; losses due to calibration factors of between 98 and 100 percent, at frequencies between 10 MHz and 1 GHz; losses due rf effective efficiencies of between 90 and 100 percent, at frequencies of above 1 GHz and up to 10 GHz; and dc errors of ±2 percent, the power indicated on the POWER dial shall be within the limits specified in the table below:

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Freq (GHz)	*Reference Power (mW)	Power dial indication (mw)
.01	0.3	0.286 - 0.308
1	0.3	
2	1	0.877 - 1.03
4	1	
7	1	
3	3	2.866 - 3.08
5	3	
8	3	
8.2	10	8.775 - 10.3
9	10	
9.8	10	

* Reference power measuring equipment shall be calibrated so that the error is no greater than ± 1/2 of 1 percent.

3.7.6 Bolometer DT-255( )/USM-161. When tested in accordance with 4.7.6, the bolometer shall meet the requirements below:

3.7.6.1 Calibration factor. The calibration factor which includes VSWR and rf effective efficiency shall be greater than 98 percent from 100 MHz to 1 GHz, when tested in accordance with the procedures of 4.7.6.1.

3.7.6.2 RF effective efficiency. The rf effective efficiency of the bolometers shall be greater than 90 percent from 10 MHz to 10 GHz, when tested in accordance with the procedures of 4.7.6.2.

3.7.6.2.1 Calibration plate. A calibration plate will be attached to the bolometer listing the rf effective efficiency value of the bolometer at 8.2, 9.0, and 9.8 MHz.

3.7.6.2.2 Swept frequency effective efficiency curve. A swept frequency curve must be provided with each bolometer. The curve must be serial numbered the same as the associated bolometer and must show a comparison between each production bolometer and a control bolometer which is traceable to NBS. The effective efficiency value of the three NBS calibration points of the control bolometer must also be shown on the X-Y swept frequency curve.

3.7.6.3 Voltage standing wave ratio. The voltage standing wave ratio (VSWR) at 9.2 GHz shall be less than 1.5, when tested in accordance with 4.7.6.3.

3.7.6.4 Temperature compensation. When the bolometer is tested in accordance with 4.7.6.4, the difference between the initial and final unbalance voltage, expressed in millivolts, shall be less than 3/4 of the difference between the initial and final temperature.

3.7.7 Attenuators.

3.7.7.1 Attenuators, Fixed CN-844( )/USM-161. When tested in accordance with 4.7.7.1, the attenuator shall meet the following requirements.

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a. Attenuation of 7 ±1.0 dB over a frequency range of 1 to 10 GHz.

b. The VSWR at 1 GHz shall be less than 1.4 measured from each end.

c. Power capacity of 5 watts for one hour shall not affect the attenuation or VSWR characteristic.

d. Calibration plate shall be attached to the attenuator indicating the actual attenuation values at the frequencies measured.

3.7.7.2 Attenuator, Fixed CN-845( )/USM-161. When tested in accordance with 4.7.7.2, the attenuators shall meet the following requirements:

a. Attenuation of 10 ±1 dB over frequency range of 1 to 10 GHz.

b. The VSWR at 1 GHz shall be less than 1.4 measured at each end.

c. Calibration plates attached to the attenuators shall indicate actual attenuation values at the frequencies measured.

3.7.8 Cable Assembly CG-2514( )/USM-161. When tested in accordance with 4.7.8, the cable assembly shall meet the following requirements:

3.7.8.1 Dielectric strength. The dielectric between the inner and outer conductors including the connectors, shall be capable of withstanding a potential of 750 volts (rms) 60 Hz for a period of one minute without arcing or other indication of breakdown.

3.7.8.2 Insulation resistance. The insulation resistance of the cable assembly shall be not less than 1000 megohms.

3.7.8.3 Continuity. Electrical continuity shall exist between the two ends of the inner and outer conductors (separately but not between) of the cable assembly, including connectors (terminations).

3.8 Service conditions. The Radio Frequency Power Test Set AN/USM-161( ) shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered; no moveable part or control shall be shifted in setting, position or adjustment; and no permanent degradation shall be caused in performance below that specified in the following service conditions:

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3.8.1 Low temperature. The test set shall be subjected to the low temperature extremes of -4°F operating and -65°F storage and shall meet the electrical requirements of 3.7.4 at times specified when tested in accordance with 4.8.1.

3.8.2 High temperature. The test set shall be subjected to the high temperature extremes of 125°F operating and 160°F storage and shall meet the electrical requirements of 3.7.4 at times specified when tested in accordance with 4.8.2.

3.8.3 Immersion. The case shall show no evidence of leakage after immersion when tested in accordance with 4.8.3.

3.8.4 Shock; bench-handling. The test set shall show no evidence of physical damage except minor surface abrasion and shall meet the electrical requirements of 3.7.4 when tested in accordance with 4.8.4.

3.8.5 Humidity. The test set shall exhibit no physical damage of parts and material, such as corrosion, rust, blistering, swelling, deterioration, etc; or electrical degradation, and shall meet the electrical requirements of 3.7.4 at times specified, when tested in accordance with 4.8.5.

3.8.6 Fungus. The test set shall show no evidence of viable fungus and shall meet the electrical requirements of 3.7.4 when tested in accordance with 4.8.6.

3.9 Interchangeability. Like units, assemblies, sub-assemblies and replaceable parts shall meet Requirement 7 of MIL-STD-454. Inspection shall be in accordance with 4.9.

3.10 Workmanship. The Radio Frequency Power Test Set AN/USM-161( ) shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454, and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives.
b. Cables, cords, and wires, electrical.
c. Wire (harnessing).
d. Lubricants
e. Plastic material and parts.
f. Rubber.
g. Tropicalization.
h. Wiring and cabling.

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i. Connectors (electrical).
j. Printed wiring assemblies.
k. Screws, other threaded devices, and related parts.
l. Tightening of threaded parts.
m. Riveting.
n. Brazing, soldering, and welding.
o. Welding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (does not include preparation for delivery) (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.12).

4.3 First article inspection. This inspection shall be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B and C inspections (see Tables I, II and III, respectively) on the three (3) samples of the completely assembled Test Set AN/USM-161( ). The first article samples shall be inspected in this order: Group A and B for all units; and designated units for specific Group C inspections. Designation of units to specific Group C inspections shall be determined by the Government representative. After completion of all Group C tests, conforming units must be reinspected and pass all Group A inspections.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery; with visual and mechanical inspection in accordance with Table I:

Item	Where required
Parts, materials and processes-general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.10

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedure and examine the contractor's inspection records. In addition, the Government--at its discretion-- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit produced shall be inspected by the contractor for conformance to all Group A requirements and tests in Table I. Each lot shall be subject to contractor and Government sampling inspection, utilizing the procedures of MIL-STD-105 using the general inspection levels, and the AQL's indicated in Table I.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

4.5.1.2 Lot size. Lot sizes shall be determined in accordance with the monthly production schedule as follows:

Production Rate	Size of Lot
500 or more per month	One week's production
51 to 499 per month	Two week's production
0 to 50 per month	One month's production

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Table I - Group A Inspection

Inspection	Req Para.	Test Para.	AQL Major	AQL Minor
Visual and mechanical				
Construction	3.2			
Parts, materials and processes-general	3.3	4.10	4.0 dphu* for the group	10 dphu for the group
Finish, protective	3.4			
Marking	3.5			
Workmanship	3.10			
Electrical				
Power requirements				
Power supply	3.7.1.1	4.7.1.1		
Bias supply	3.7.1.2	4.7.1.2		
Calibration				**
Constant current supply	3.7.2.1	4.7.2.1		
Automatic zero adjustment	3.7.2.2	4.7.2.2		
Maximum power calibration	3.7.2.3	4.7.2.3		
Minimum power calibration	3.7.2.4	4.7.2.4	2.5 dphu combined for the group	
Calibration verification				
Power meter	3.7.3.1	4.7.3.1		
Comp attenuator	3.7.3.2	4.7.3.2		
Operational	3.7.4	4.7.4		
Accuracy of power measurement	3.7.5	4.7.5		
Bolometer DT-255( )/USM-161			2.5 dphu combined for the group	**
Calibration factor	3.7.6.1	4.7.6.1		
RF effective efficiency	3.7.6.2	4.7.6.2		
Calibration plate	3.7.6.2.1	4.7.6.2.1		
Swept frequency efficiency curve	3.7.6.2.2	4.7.6.2.2		
Voltage standing wave ratio	3.7.6.3	4.7.6.3		
Temperature compensation	3.7.6.4	4.7.6.4		
Attenuators				
Attenuator, Fixed CN-844( )/USM-161	3.7.7.1	4.7.7.1	1.0 dphu combined for the group	**
Attenuator, Fixed CN-845( )/USM-161	3.7.7.2	4.7.7.2		

* Defects per hundred units.
** All electrical tests are major.

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4.5.2 Group B inspection. This inspection , including contractor and Government sampling inspection, shall conform to Table II and to the special procedures for small inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection.

4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group B Inspection

Inspection	Req Para	Test Para	AQL
Interchangeability	3.9	4.9	6.5 dphu*
Cable Assembly CG-2514( )/U			
Dielectric strength	3.7.7.1	4.7.7.1	6.5 dphu for the group
Insulation resistance	3.7.7.2	4.7.7.2	
Continuity	3.7.8.3	4.7.8.3	

* Defects per hundred units.

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table III and shall be performed on sample units that have been subjected to and met all Group A and B tests. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for inspection.

4.5.3.1.1 Four (4) samples of complete equipment shall be selected at random, one (1) sample for each Group 1 through Group 4 in Table III. The samples selected shall be at the start of the contract from the first quality conformance inspection lot. These samples shall constitute Group C requirement for first 100 units produced.

4.5.3.1.2 Thereafter, one (1) sample of the complete equipment shall be selected at random for Group 1 and Group 2 in Table III from every 100 units, and fraction thereof produced.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

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Table III - Group C Inspection

Inspection	Req Para	Test Para
Group 1		
Low temperature	3.8.1	4.8.1
High temperature	3.8.2	4.8.2
Immersion	3.8.3	4.8.3
Group 2		
Shock, bench-handling	3.8.4	4.8.4
Humidity	3.8.5	4.8.5
Group 3		
Fungus	3.8.6	4.8.6
Group 4		
Construction	3.2	4.11

4.5.3.3 Evaluation of samples submitted to individual Group C inspection. After referenced samples in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test, unless otherwise specified in the individual test, the sample shall meet the requirements of 3.7.4. when tested in accordance with 4.7.4.

4.5.3.4 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Government quality assurance representative. After corrective action has been taken, additional samples shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government). The final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

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4.5.4 Reinspection of conforming Group B and C samples. Unless otherwise specified, samples which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or purchase order provided all damage is repaired and the samples are subjected to and pass Group A inspection.

4.5.5 Disposition of Group C samples. Unless otherwise specified, samples which have been subjected to and passed Group C inspection may be delivered on the contract or purchase order provided samples meet requirements of 4.5.4.

4.6 Preconditioning.

4.6.1 Bounce preconditioning. Each equipment produced, with all units mounted in the case as for field transportation, shall be placed in its normal operating position on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.6.1.

4.6.2 Electrical preconditioning. Each equipment produced shall be placed in its normal operating position on a test table (or in a test rack) with room temperature at 77° ± 5°F, with specified operating voltage (115 Vac) applied from a suitable power source. With all controls in the normal operating position, the equipment shall be turned on and remain on for a continuous 24 hour period and monitored. In case of failure, the equipment shall be repaired, recalibrated, and the electrical preconditioning test repeated for the failed equipment. A failed item analysis report shall be submitted on each failure with contents of report in accordance with 5.10.5 of MIL-STD-781. Each equipment shall meet the requirements of 3.6.2.

4.7 Electrical. Figure 1 indicates the test equipment recommended for the calibration and testing of the electrical and operational characteristics of the test set. Where manufacturer's commercial items are indicated, any equivalent type test equipment, or method, approved by the Government production engineer, may be used to demonstrate compliance with specification requirements.

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4.7.1 Power requirements. When tested in accordance with the following procedures, the equipment shall meet the requirements of 3.7.1.

4.7.1.1 Power supply. Connect the test set, with bolometer attached, to the adjustable line transformer and proceed as below:

a. Set transformer output to 115V, 60 Hz, the POWER RANGE switch of the test set to 10 mW and the THERM RES to 200 ohms. Measure the B negative voltage. Repeat above with line voltage frequency of 400 Hz.

b. Vary the line voltage transformer output from 105 to 129 volts, 60 Hz continuously checking the B negative voltage. Repeat with line voltage frequency of 400 Hz.

c. Measure the B negative ripple with an oscilloscope at 115V, 60 and 400 Hz respectively.

4.7.1.2 Bias supply. When tested in accordance with the following, the equipment, with bolometer attached, shall meet the requirements of 3.7.1.2.

a. Measure gain of bias oscillator, using a dual input oscilloscope with a preamplifier, (or equivalent).

b. Connect the bridge input (pins 2 and 7 of V8) to channels A and B respectively.

c. Connect the bridge output (junction of R85 and R33) to the trigger or external sweep input.

d. Adjust the oscilloscope controls for an indication on the display screen of an ellipse at about a 45 degree angle.

e. Calculate the vertical and horizontal deflections from the oscilloscope gain control settings.

f. Calculate gain by dividing the horizontal deflection voltage by the vertical deflection voltage.

4.7.2 Calibration. (The bolometer mount is not attached to the power meter when performing any of the following calibration procedures).

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4.7.2.1 Constant current supply. The constant current supply, when calibrated in accordance with the following procedures, shall meet the requirements of 3.7.2.1.

a. Set the LINE ON switch to ON and allow a 15-minute warmup period.

b. Set the POWER RANGE switch to 1 MW, 0 DBM.

c. Rotate the COMP ATTENUATOR control fully counterclockwise.

d. Connect the positive lead of the digital voltmeter to pin 7 of tube V5.

e. Connect the negative lead of the voltmeter to the negative side of capacitor C1B (junction of capacitor C1B and resistor R14).

f. Adjust potentiometer R13 for an indication of 50 volts on the digital voltmeter.

g. Remove the positive lead of the voltmeter and reconnect it to pin 6 of tube V7. Adjust potentiometer R21 for an indication of 100 volts on the voltmeter.

h. Repeat the procedure given in d, e, and f above.

i. Remove the positive lead of the voltmeter and reconnect it to pin 6 of tube V7.

j. Adjust the COMP ATTENUATOR control for an indication of 69.003 volts on the voltmeter.

k. Loosen the two setscrews on the COMP ATTENUATOR control, and rotate the dial (without rotating the shaft) until the zero line on the dial is under the hairline.

l. Tighten the two setscrews.

m. If the indication of the voltmeter is not 69.003 volts, repeat the procedure given in j, k and l above.

n. Insert plug PJ-047 into jack J2. Remove the voltmeter test leads.

o. Set the THERM RES switch to 100 ohms.

p. Connect the positive lead of the voltmeter to chassis ground

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q. Connect the negative lead of the voltmeter to the junction of R72 and R73.

r. Adjust potentiometer R13 for an indication of 2.5028 volts on the voltmeter.

s. Adjust the COMP ATTENUATOR control first for +2.2 dB and then for -2.2 dB. The voltmeter readings shall be within the limits specified in 3.7.2.1. With the COMP ATTENUATOR set for $^{0}dB_{set}$ POWER RANGE switch for 0.3, 3 and 10 mW; read voltmeter at each range. The voltages shall be within the limits specified in 3.7.2.1.

t. Disconnect equipment. Remove plug PJ-047 from jack J2.

4.7.2.2. Automatic zero adjustment. When calibrated in accordance with the following, the voltage (E_1) measured across C22 shall meet the requirements of 3.7.2.2.

a. Perform the procedures described in 4.7.2.1 above.

b. Insert plug PJ-051 into jack J1.

c. Set the POWER dial to 0.

d. Set the BIAS-READ switch to READ.

e. Connect the voltmeter across capacitor C22 (positive test lead to chassis ground) and note the indication. This is E_1.

f. Set the BIAS-READ switch to BIAS and adjust potentiometer R75 until the indication on the voltmeter is equal to E_1.

4.7.2.3 Maximum power calibration. When calibrated in accordance with the following, the voltage E_3 shall meet the requirements of 3.7.2.3.

a. Perform the procedures described in 4.7.2.2 above.

b. Set the BIAS-READ switch to READ.

c. Set the POWER control dial to 1 (on the outermost scale).

d. Note the indication on the voltmeter. This is E_2.

e. Using the following formula, solve for E_3: $E_3 = 0.15645 (E_1 - E_2)$.

f. Adjust potentiometer R74 until the indication on the voltmeter is equal to E_3.

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4.7.2.4 Minimum power calibration. When calibrated in accordance with the following, the voltage E_4 shall meet the requirements of 3.7.2.4.

a. Perform the procedures described in 4.7.2.3 above.

b. Connect a 200-ohm, ±0.1 percent, 1/2 watt resistor from the junction of resistors R67 and R68 to chassis ground.

c. Set the THERM RES switch to 200 ohms.

d. Remove plug PJ-051 from jack J1. Insert plug PJ-047 into jack J2.

e. Set the BIAS-READ switch to READ.

f. Using the following formula, solve for E_4: $E_4 = 0.57822 (E_1 - E_2)$.

g. Adjust the potentiometer R69 until the indication on the voltmeter is equal to E_4.

h. Remove plug PJ-047 from jack J2.

i. Disconnect the 200-ohm resistor and the voltmeter.

4.7.3 Calibration verification. Calibration verification shall be performed as described below.

4.7.3.1 Power meter. When tested in accordance with the following, the meter shall meet the requirements of 3.7.3.1:

a. Connect one side of the 200-ohm precision resistor to the junction of resistors R67 and R68.

b. Connect the other side of the 200-ohm resistor to the ground side of resistor R73.

c. Connect the voltmeter across the 200-ohm resistor.

d. Turn on the voltmeter.

e. Set the controls on the power meter as follows:

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(1) LINE ON switch to ON.

(2) BIAS-READ switch to READ.

(3) THERM RES switch to 200 ohm.

(4) POWER RANGE switch to 1 MW 0 DBM.

(5) POWER control dial to 0 (on the outermost scale).

(6) COMP ATTENUATOR control to 0.

f. Measure the voltage across the resistor with the voltmeter and note the indication. Record the voltage as V_0.

g. Set the POWER control dial to 1 on the outermost scale and note the new indication on the voltmeter. Record voltage as V_1.

h. Divide the difference of the squares of the indications noted in f and g above by 0.200.

4.7.3.2 Comp attenuator. When tested in accordance with the following, the comp attenuator shall meet the requirements of 3.7.3.2.

a. Perform the procedures in 4.7.3.1 above.

b. Set POWER dial to 0 (on the outermost scale).

c. Set COMP ATTENUATOR to -2.2 dB and read the voltmeter.

d. Set COMP ATTENUATOR to +2.2 dB and read the voltmeter.

4.7.4 Operational. When tested in accordance with the following procedures, the equipment shall meet the requirements of 3.7.4.

4.7.4.1 Sensitivity of null indication.

a. Connect Bolometer to test set.

b. Control settings as follows:

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(1) LINE ON to ON.

(2) BIAS READ to READ.

(3) POWER dial to 0.10 (outermost scale).

(4) COMP ATTEN to 0.

(5) THERM RES to 200.

(6) POWER RANGE to 1 MW.

(7) BIAS COARSE for null on indicator.

(8) BIAS FINE for null on indicator.

c. Rotate POWER dial until needle on null indicator meter is at one of the inscribed lines near the null point.

d. Record indication on POWER dial (approximately 0.09 mW).

e. Rotate POWER dial until needle on null indicator is at the other inscribed line near null point.

f. Record indication on POWER dial (approximately 0.11 mW).

g. Subtract the indications derived in d. and f. above, divide the result by 2. Result shall meet requirements of 3.7.4.1.

4.7.4.2 Power measurement. Using an approved power standard (HP Calorimeter or equal) and with test set control set for measurement of power:

a. Apply RF Power of 0.3, 1 and 3 mW at 1 GHz (cw) to the Bolometer and test set.

b. Observe and record the indications on the 0.3 mw, 1 mW and 3 mW POWER RANGES. Result shall meet requirements of 3.7.4.2.

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4.7.4.3 Zero shift.

a. Set controls as follows:

(1) LINE ON to ON.

(2) BIAS READ to READ.

(3) THERM RES to 200.

(4) POWER RANGE to 0.30 MW.

(5) COMP ATTEN to 0.

(6) POWER dial to 0.10 (outermost scale).

(7) BIAS COARSE to null on indicator.

(8) BIAS FINE to null on indicator.

b. Connect the 10-dB Attenuator to the Bolometer mount.

c. Adjust POWER dial for null on null indicator.

d. Note the indication on the POWER dial. It shall meet the requirements of 3.7.4.3.

4.7.5 Accuracy of power measurement. When tested in accordance with the following, the equipment shall meet the requirements of 3.7.5.

a. Connect appropriate signal sources to the input of the power meter.

b. Monitor the input level with a standard calorimeter or other equivalent approved power standard..

c. Check the accuracy of the power measurements (using the POWER RANGE which will indicate closest to full scale) in accordance with the following input levels andfrequencies:

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(1) 0.3 mW at 10 MHz and 1 GHz.

(2) 1 mW at 2, 4, and 7 GHz.

(3) 3 mW at 3, 5 and 8 GHz.

(4) 10 mW at 8.2, 9 and 9.8 GHz.

4.7.6 Bolometer DT-255( )/USM-161. When tested in accordance with the following, the bolometer shall meet the requirements of 3.7.6.

4.7.6.1 Calibration factor. Calculate the calibration factor of the produced bolometer mount by comparing power measurements taken first using a bolometer mount traceable to the National Bureau of Standards at frequencies of 10 MHz, 100 Mhz, and 1 GHz, measuring power of 3 and 10 mW.

4.7.6.2 RF effective efficiency. The rf effective efficiency shall be tested in accordance with SM-C-429576 (Test Procedures, Bolometer) and meet the requirements of 3.7.6.2. SM-C-429576 illustrates a system block diagram for rapid efficiency and reflection coefficient testing of the bolometers. The following or equivalent is required for the measurement of rf effective efficiency:

H/P 3440A Digital Voltmeter w/3443A High Gain/Auto Range Unit
H/P K31-431B Efficiency Calibrator
H/P H01-694B Sweep Oscillator
3 each H/P 431B Power Meter
2 each H/P H852C Directional Coupler
H/P 478A 200 ohm Negative Thermister Mount
H/P H920A Adjustable Short
H/P H532A Frequency Meter
Moseley Model 135 X-Y Recorder

4.7.6.2.1 Calibration plate. The bolometer shall be checked to insure that a calibration plate is mounted on it listing the effective efficiency values specified in 3.7.6.2.1.

4.7.6.2.2 Swept frequency curve. The equipment shall be checked to insure that a swept frequency curve in accordance with the requirements of 3.7.6.2.2 is included with each equipment.

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4.7.6.3 Voltage standing wave ratio. With the equipment connected as in Figure 2, keep the null indicator zeroed and the THERM RES switch in the 200 ohm position. Measure the VSWR at 9.2 GHz. Measurement shall meet the requirements of 3.7.6.3.

4.7.6.4 Temperature compensation. Connect the bolometer in accordance with the circuit shown in Figure 3. With the switch in the ADJUST position, null the indicating meter. Switch to the MATCH position and record the unbalance voltage. Also record the temperature of the bolometer in degrees fahrenheit. Raise the temperature of the bolometer to reach equilibrium. Switch to ADJUST position and restore the meter to null indication. Switch to MATCH position and record the unbalance voltage and the exact temperature of the bolometer in degrees fahrenheit. Results shall be in accordance with the requirements of 3.7.6.4.

4.7.7 Attenuators.

4.7.7.1 Attenuator, Fixed CN-844( )/USM-161. The attenuator shall meet the requirements of 3.7.7.1 when tested in accordance with the following:

a. Using the test set-up of Figure 4, or equivalent, measure the attenuation at 1, 2, 4, 7 and 10 GHz and VSWR at 1 GHz.

b. Apply 5 watts to the attenuator for one hour. Recheck attenuation and VSWR at 1 GHz.

4.7.7.2 Attenuator, Fixed CN-845( )/USM-161. The attenuator shall meet the requirements of 3.7.7.2 when tested in accordance with the following:

a. Using the test set-up of Figure 4, or equivalent, measure attenuation at 1, 1.5, 2, 5 and 10 GHz and VSWR at 1 GHz.

b. Apply 5 watts to the attenuator for one hour. Recheck attenuator and VSWR at 1 GHz.

4.7.8 Cable Assembly CG-2514( )/USM-161. The cable assembly shall meet the requirements of 3.7.8 when tested in accordance with the following procedures:

4.7.8.1 Dielectric strength. The assembled cable shall be tested for compliance with the requirements of 3.7.8.1 when a potential of 750 volts (rms) 60 hz is applied to the terminated end of the cable.

4.7.8.2 Insulation resistance. The characteristic of the cable assembly shall be measured for compliance with 3.7.8.2, using a suitable megohmeter.

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4.7.8.3 Continuity. The continuity of the inner and outer conductors, including terminations, shall be tested using a suitable ohmmeter and shall meet the requirements of 3.7.8.3.

4.8 Service conditions. The Radio Frequency Power Test Set AN/USM-161( ) shall be tested in accordance with the following service conditions and meet the requirements of 3.8 in addition to any of the electrical requirements specified.

4.8.1 Low temperature. The test set in its normal operating configuration shall be subjected to the low temperature test specified in Procedure I, Method 502 of MIL-STD-810, with temperature in Steps 2 and 4 maintained for 24 hours, and shall meet the requirements of 3.8.1.

4.8.2 High temperature. The test set in its normal operating configuration shall be subjected to the high temperature test specified in Procedure I, Method 501 of MIL-STD-810 with temperature in Step 4 held for 24 hours, and shall meet the requirements of 3.8.2.

4.8.3 Immersion. The test set transit case, less the unit, shall be braced (dummy front panel and chassis) and weighted to simulate the weight of the equipment shall be subjected to the immersion test specified in Procedure I, Method 512 of MIL-STD-810, except the temperature of the water shall be 70° ± 5°F and the test item shall be 110° ± 5°F, and shall meet the requirements of 3.8.3.

4.8.4 Shock, bench handling. The test set shall be subjected to the bench handling shock test specified in Procedure V, Method 516 of MIL-STD-810, except 30 degree instead of 45 degree angle drop, and shall meet the requirements of 3.8.4.

4.8.5 Humidity. The test set, in its normal operating configuration shall be subjected to the humidity test specified in Procedure II, Method 507 of MIL-STD-810, and shall meet the requirements of 3.8.5.

4.8.6 Fungus. The test set, in its normal operating configuration shall be subjected to 28 days of the fungus test specified in Procedure I, Method 508 of MIL-STD-810 and shall meet the requirements of 3.8.6.

4.9 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.9 . When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

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b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers and mountings.

d. Location of switches, sockets and fasteners.

e. Location of cover and fasteners on carrying cases which must mate.

f. Location of panel mounting screw holes on basic unit for mating with case.

4.10 Visual and mechanical. The test set shall be subjected to a thorough visual and mechanical inspection. The defects shall be classified in accordance with MIL-STD-252.

4.11 Construction. The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on contract against the drawings cited for construction in 3.2. The EVR shall consist of the following:

4.11.1 An audit to establish that as built equipment (unit, assembly, module, or part) is in accordance with the cited end product drawings.

4.11.2 An audit to establish that as built configuration of a unit, assembly, module, or part meets the acceptance requirements (other than equipment specification performance) specified for each unit, assembly, module, or part. The audit shall record all acceptance test methods used together with resulting verification test data.

4.11.3 The contractor shall prepare an EVR report containing the following information:

a. Identification of unit, assembly, module, or part and details shall be in accordance with 4.11.1 and 4.11.2.

b. Discrepancies noted.

c. Corrective action taken.

4.11.4 The contractor shall furnish the Government inspector a copy of the EVR report no later than fifteen (15) days prior to the submission of the first production unit or lot. The results of the EVR shall be subjected to inspection by Government personnel at the time the first unit or production lot is offered for acceptance, Government inspection will be to the depth and extent necessary to demonstrate that the as built hardware is in accordance with the cited drawings. The contractor shall provide the following:

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a. Segregation of units, assemblies, modules, and parts to permit Government personnel access for detailed inspection.

b. As required, responsible personnel from each functional department available for discussions in their respective areas.

c. Adequate administrative support for the EVR.

4.12 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. The Radio Frequency Power Test Set AN/USM-161( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. The Radio Frequency Power Test Set AN/USM-161( ) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged Method 1C-1.

5.1.1.4.2 Electron tubes. All tubes not securely clamped in place shall be removed and with spare tubes shall be individually packaged in accordance with MIL-E-75.

5.1.1.4.3 Fuses. All fuses shall be packaged in accordance with Method III as follows: Cushion the fuses by wrapping together in paperboard, conforming to PPP-P-291, Type I. Secure wrap with gummed paper tape conforming to PPP-T-45, Type III, Grade A.

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5.1.1.4.4 Lamp. Each lamp shall be individually packaged Method III as follows: Cushion each lamp by wrapping in paperboard conforming to PPP-P-291, Type I. Secure wrap with gummed paper tape conforming to PPP-T-45, Type III, Grade A.

5.1.1.4.5 Cable. Each cable shall be packaged Method III as follows: Wind each cable into a coil of proportionate dimensions and tie in three places with cotton twine conforming to T-T-871.

5.1.1.4.6 Stowage. Stow within the compartments of the case items packaged as specified in 5.1.1.4.2 thru 5 .1.1.4.5. Fill all voids with paperboard conforming to PPP-P-291, Type I, to prevent movement.

5.1.1.4.7 Attenuators and bolometer. Place the attenuators and bolometer in the clips provided in the transit case cover, close the cover and secure with fasteners provided.

5.1.1.4.8 Radio Frequency Power Test Set AN/USM-161( ). Each test set shall be individually packaged Method III as follows: Cushion each test set on all surfaces with cells or pads or both conforming to PPP-F-320, W5c, weather-resistance designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5c, weather-resistant. Place the technical literature packaged as specified in 5.1.1.4.1 in the box directly under the lid of the box. Box closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. The Radio Frequency Power Test Set AN/USM-161, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited in-transit storage.

5.2 Packing. Packing shall be level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container or its contents (see 6.2b).

5.2.1 Level A.

5.2.1.1 Palletized load. A quantity of Radio Frequency Power Test Set AN/USM-161( ) packaged as specified in 5.1, shall be placed on a pallet, load Type I, conforming to MIL-STD-147, except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides

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extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I class B. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Palletized load. A quantity of Radio Frequency Power Test Sets AN/USM-161( ), packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. When quantities per destinations are less than a pallet load, a quantity of Radio Frequency Power Test Sets AN/USM-161( ), packaged as specified in 5.1, shall be placed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet mounted on pallets conforming to NN-P-71, type N. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by nonmetallic strapping or pressure-sensitive filament tape banding conforming to PPP-S-760, type II and PPP-T-97, type IV respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

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5.2.3 Level C.

5.2.3.1 Palletized load. A quantity of Radio Frequency Power Test Sets AN/USM-161( ) packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard cap shall be class domestic.

5.2.3.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of Radio Frequency Power Test Sets AN/USM-161( ) packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Radio Frequency Power Test Set AN/USM-161( ) is intended for use at direct and general support and depot maintenance. It is an ac operated, temperature-compensated portable test unit used to measure radio frequency power over a frequency range of 10 MHz to 10 GHz, from either continuous wave (cw), modulated, or pulsed sources.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level of preservation and packaging and packing (see section 5).

c. When rough handling and functional tests are required.

d. Quantity of first articles required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Marking for shipment and shipping containers.

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6.3 Nomenclature. The contractor should apply the nomenclature (item name and type designation) in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design for example: SG-12( )/U to SG-12F/U, or AN/URM-73( ) to AN/URM-73C, or AN/USM-161( ) to AN/USM-161A.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.4).

6.5 Verification inspection. Verification inspection by the Government will be limited to the amount deemed necessary to determine compliance with the contract or purchase order and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of sampling inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodians:
Army - EL

Preparing Activity:
Army-EL

User Activities:

Project Number: 6625-A459

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Description	Mfgr.	Model
Digital VTVM	John Fluke	801
Oscilloscope	Hewlett-Packard	130B
Variable Transformer	Powerstat	116
200 ohm - 1% Precision Resistor, 1/2 watt	--	
Connector	MS310E-14S-15P	
AC Voltmeter	Simpson	260
Plug	Switchcraft	100A
Plug	Switchcraft	440
Signal Source	FXR	L-772A
Signal Source	FXR	S-772A
Signal Source	FXR	C-772A
Signal Source	Varian	X-13
Power Supply	PRD	809
Slotted Section Carriage	Hewlett-Packard	809B
Slotted Section Body	Hewlett-Packard	806B
Slotted Section	Gen. Microwave Corp.	N200
Standard Bolometer Mount	* Gen. Microwave Corp.	N401 with 408 element
Frequency Meter	Hewlett-Packard	X532B
Tuner	PRD	306/A
Standing Wave Amplifier	Hewlett-Packard	415B
Attenuator Calibrator	Weinschel	BA-5
Coax to Waveguide Adapter	PRD	354B
Level Set (Waveguide)	PRD	159B
Standard Attenuator	*CN-844( )/USM-161	
Standard Attenuator	*CN-345( )/USM-161	
Calorimeter	Hewlett-Packard	434A

* These items shall be "first production" units, calibrated, supported by calibration and test data, submitted to the Government and approved for use as "Test Standards."

The equipment required for the measurement of rf effective efficiency is listed in paragraph 4.7.5.2.

Figure 1 - Test Equipment Required

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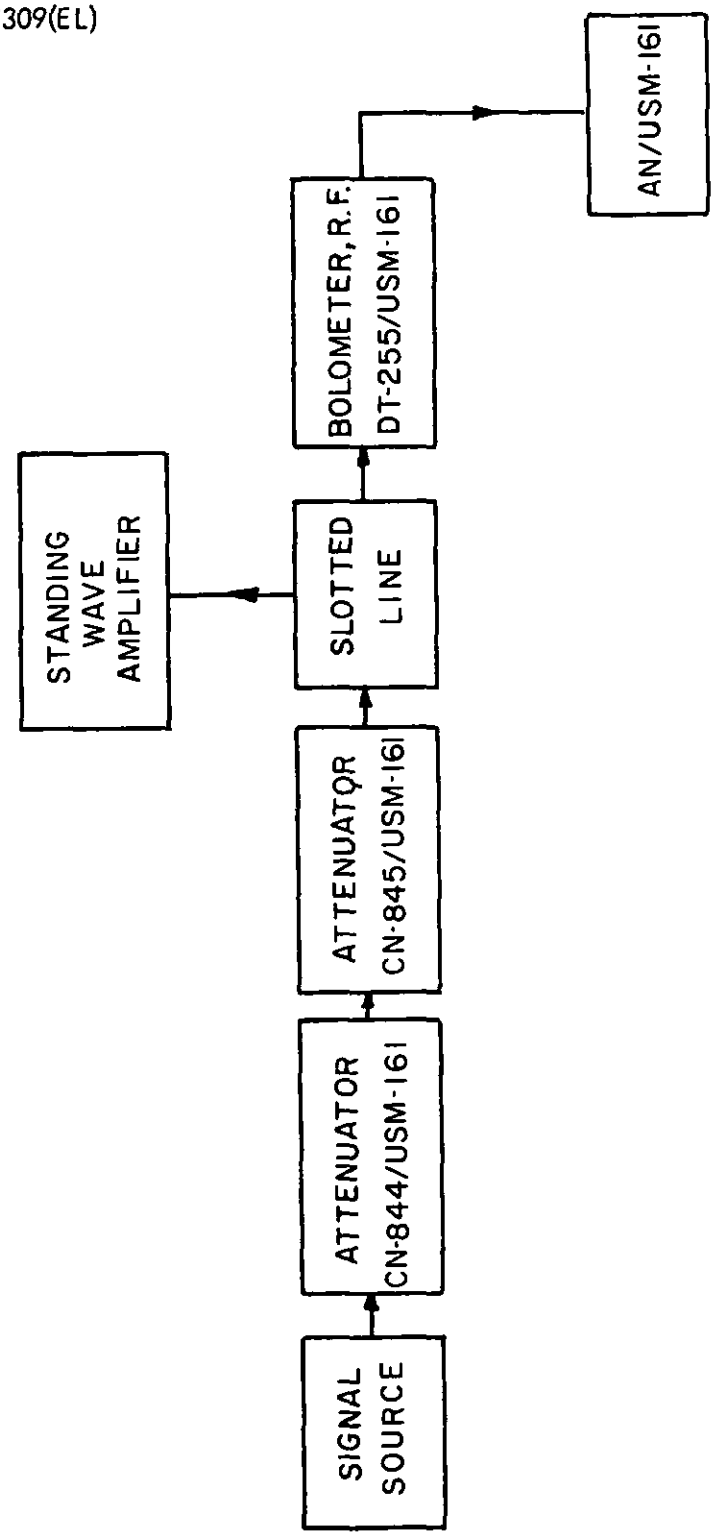


Figure 2 - VSWR Measurement, Bolometer, RF DT-255/USM-161

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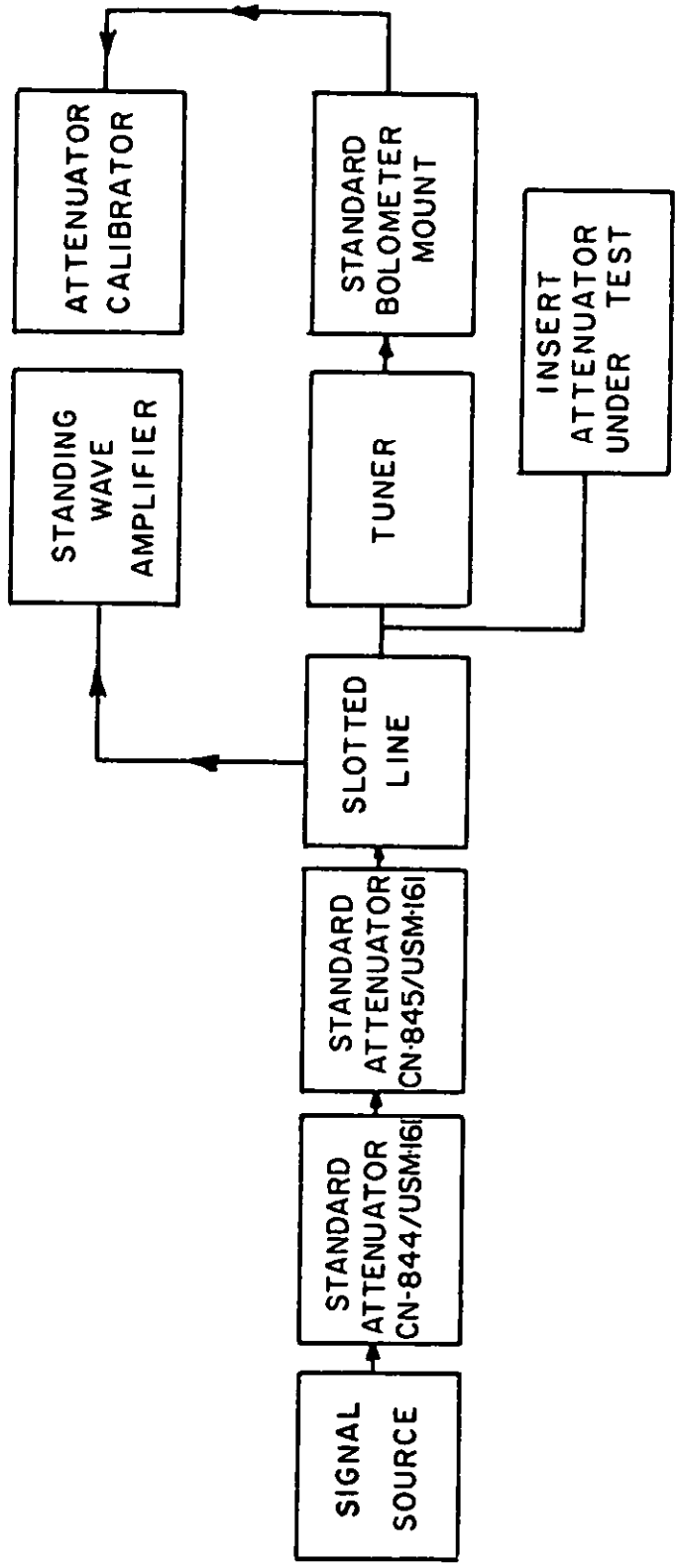


Figure 4 - VSWR and attenuation measurement of Attenuators CN-844/USM-161 and CN-845/USM-161

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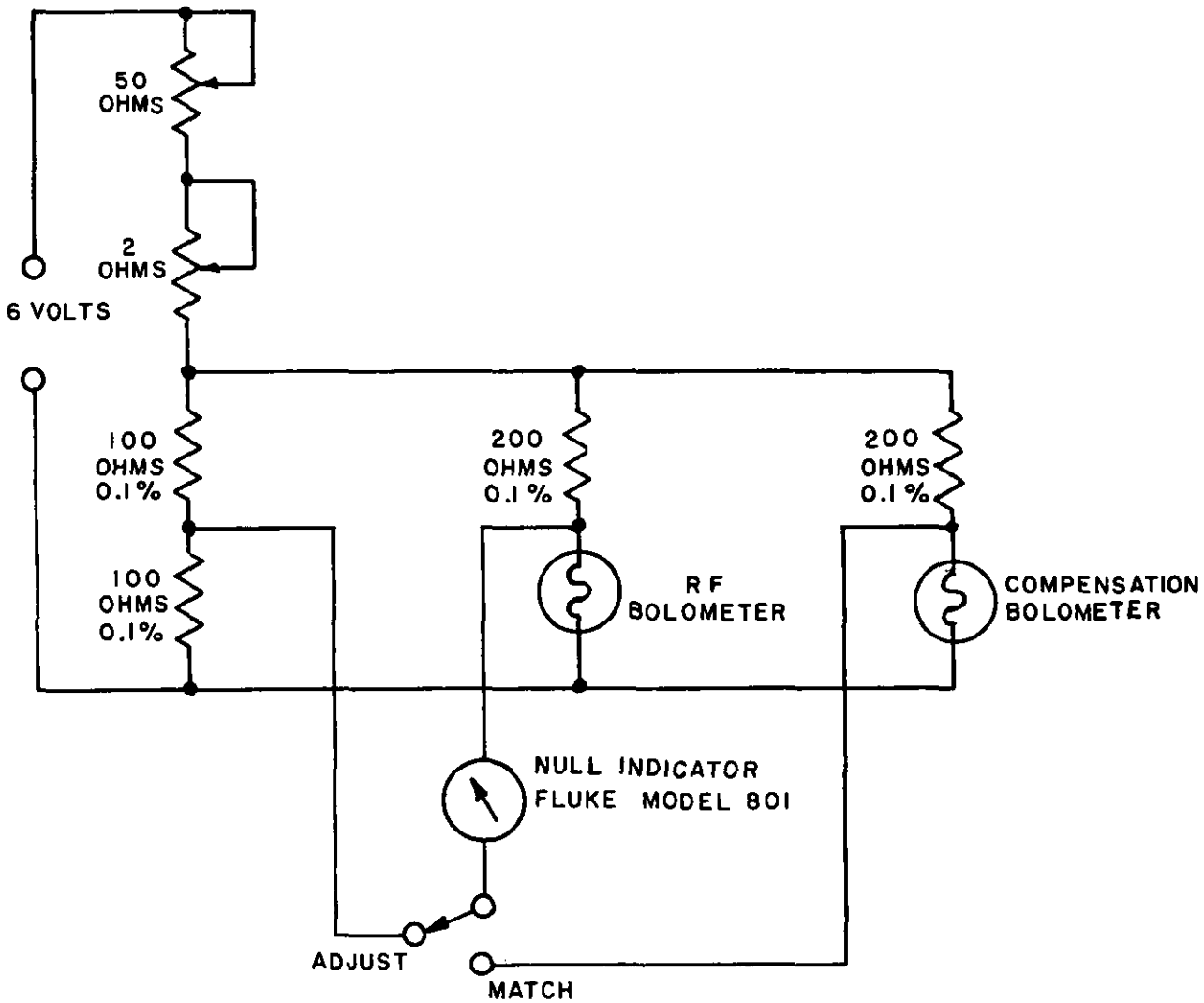


Figure 3 - Temperature Compensation of Bolometer, RF DT-255/USM-161

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commanding General
U. S. Army Electronics Command
Attn: AMSEL-TD-SS
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4663, WHICH IS OBSOLETE C-8279

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